V O L T S L O T

VoltSlot

EV Charging Station Management and Slot Reservation System

User Manual

Student name	[Enter student name]
USN	[Enter USN]
Department	[Enter department]
College	[Enter college name]
Project guide	[Enter guide name]
Academic year	[Enter academic year]

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Contents

Customer Guide

Administrator Guide

Screenshot Insertion Points

Account and Safety Notes

Customer Guide

Follow these steps to find a station, reserve an available slot, and manage your booking.

1. Register and sign in

1. Open VoltSlot and choose customer registration, or choose the User Portal sign-in link.
2. Enter your name, email, and a password meeting the form rules (8–72 characters/bytes including uppercase, lowercase, and a number).
3. After registration, VoltSlot signs in the new customer. On later visits, sign in with your email and password.

2. Browse charging stations

1. Use the public landing-page directory or select *Find stations* in the User Portal.
2. Search by name/address/city. Narrow results by city, connector type, slot availability, or station status.
3. Open a station card's *View details* link to review location, price, opening hours, connectors, and the slot schedule.

3. Reserve a slot

1. Sign in to a customer account. Guests can browse schedules but must sign in before reserving.
2. Select an available future slot and choose *Reserve*.
3. Enter the estimated energy in kWh, then select *Confirm reservation*.
4. Review the confirmation and estimated total. If another user claims the slot first, refresh and select another available time.

4. View and cancel bookings

1. Choose *My bookings* in the User Portal.
2. Review the booking's station, schedule, energy/price, and status.
3. For an eligible confirmed booking that has not started, choose *Cancel booking* and confirm. The application refreshes the list; the slot is released when the cancellation succeeds.

5. Update profile / sign out

Choose *My profile* to update the account name or email address. Choose *Sign out* to end the browser session.

Administrator Guide

The administrator portal provides station, slot, booking, and analytics tools.

1. Sign in

Use the Admin Portal sign-in entry and an existing admin account. Public account registration is limited to customer accounts. The backend repository has a seed script for the first admin and expects ADMIN_EMAIL and ADMIN_PASSWORD in a local backend environment file; the seed script enforces input checks and refuses an existing account. Do not share the file or its values.

2. Review dashboard and analytics

The dashboard shows station and slot totals, booking status counts, bookings-by-status chart, most-booked station chart, and recent booking activity. Cards and shortcuts link to station/slot and booking operations.

3. Manage stations

1. Choose *Stations & slots* in the admin navigation.
2. Select *Add station* to enter a name, address/city, longitude/latitude, one or more connectors, price/kWh, operating hours, and operational status.
3. Select *Edit* on an existing row to update its details; save or cancel.
4. Choose delete and confirm only when appropriate. A station with dependent slots or bookings cannot be deleted.

4. Manage slots

1. Choose *Manage slots* for a station.
2. Choose *Add slot*; enter the scheduled start and end and create it.
3. Use Edit to adjust an eligible slot. Use Activate/Deactivate for availability status or Delete for a slot without booking history.
4. Booked slots and slots referenced by booking history are protected from conflicting schedule changes/deletion.

5. Manage bookings

1. Choose *Manage bookings*.
2. Filter the list by all, confirmed, cancelled, or completed status.
3. Cancel a confirmed eligible booking if needed, or mark a confirmed booking complete after the scheduled slot has ended.

6. Sign out

Use the Sign out control in the portal navigation when administration work is complete.

Screenshot Insertion Points

Screenshots were not captured for this documentation build because no isolated demo account/database was established for safe browser interaction. These labelled boxes are intentionally not presented as application screenshots.

[SCREENSHOT PLACEHOLDER: Landing page and station listing]
Insert a genuine capture from a running VoltSlot build.

[SCREENSHOT PLACEHOLDER: User dashboard]
Insert a genuine capture using a test customer account.

[SCREENSHOT PLACEHOLDER: Slot reservation and confirmation]
Insert a genuine capture using a disposable test slot/account.

[SCREENSHOT PLACEHOLDER: My Bookings]
Insert a genuine capture with private details redacted.

[SCREENSHOT PLACEHOLDER: Admin dashboard]
Insert a genuine capture using a test administrator account.

[SCREENSHOT PLACEHOLDER: Station and slot management]
Insert a genuine capture using a safe test environment.

Account and Safety Notes

- Only register ordinary customer accounts through the public registration form. Administrator provisioning is handled separately.
- Do not share account passwords, JWTs, MongoDB connection strings, environment files, or customer records in screenshots.
- Booking creation is restricted to customer-role tokens by the API; admin UI hiding is supplementary and not the authorization boundary.
- Errors can mean the slot was reserved by someone else, is no longer eligible, or is blocked by existing booking history. Review the message and refresh before retrying.
- The current model protects a slot ID from a second confirmed booking, but does not define station-level capacity for distinct overlapping slot records.
- For deployment/configuration trouble, contact the system administrator; this guide does not describe production deployment.